

- 3 A cantilever spar cable-stayed bridge is a unique yet functional variation of the traditional cable suspension bridge.

In one such model bridge, a bridge beam is supported by a cable. The cable connected at an angle of 30° to a non-uniform cantilever spar of length L slanted at an angle of 50° from the ground with a base at P, as shown in Fig. 3.1.

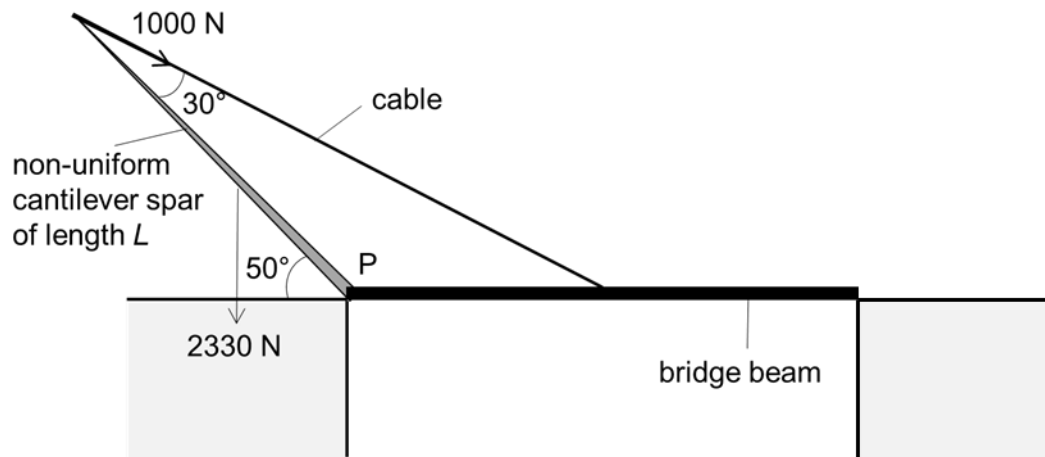


Fig 3.1 (not to scale)

The tension in the cable is 1000 N and the weight of the cantilever spar is 2330 N.

- (a) By taking moments about P, show that the centre of mass of the cantilever spar is located at a distance of $0.33L$ from P.

[3]

Solution:

Taking moments about P:

Let d be the distance of the centre of mass from P

Moment due to the tension = Moment due to the weight

$$T \times L \sin(30) = W \times d \cos(50)$$

$$1000 \times 0.5L = 2330 \times 0.643d$$

$$d = 0.33L \text{ (shown)}$$

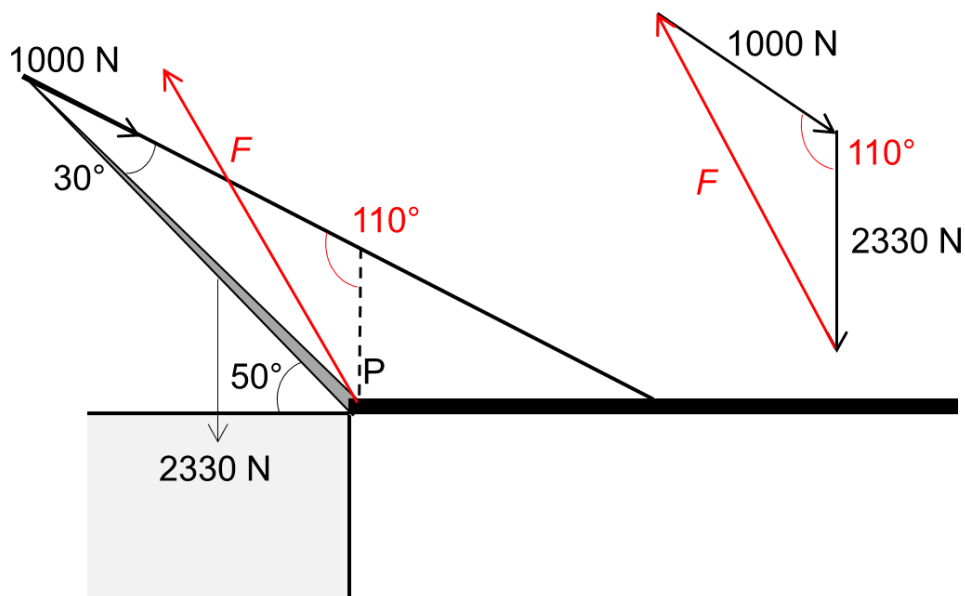
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- (b) Calculate the magnitude of the force acting on the cantilever spar at P.

magnitude of force at P = N [3]

Solution:

Since the forces are in equilibrium, the three forces must form a closed vector triangle.



To solve for F, use the cosine rule:

$$F^2 = 1000^2 + 2330^2 - 2(1000)(2330)\cos 110$$

$$F = 2830 \text{ N}$$

(c) The traditional cable-stayed bridge design is shown in Fig 3.2

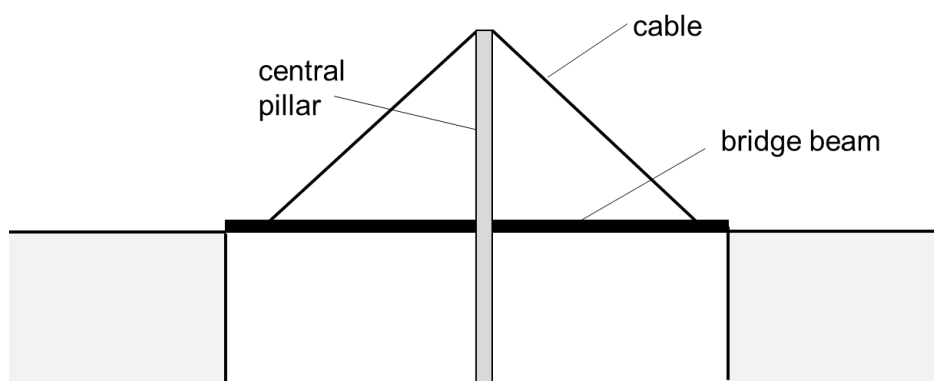


		Fig 3.2 Suggest, with a reason, one advantage a cantilever spar cable-stayed bridge may have over a traditional cable-stayed bridge design. [2]	
		Solution: the cantilever spar requires fewer cables to support a bridge since there is no need to support spans on either side of the central pillar OR wider space of vessels to pass through under the bridge since there is no central pillar	1 1 1 1

4	(a)	A battery of electromotive force (e.m.f.) 0.0 V and internal resistance r is connected to two
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